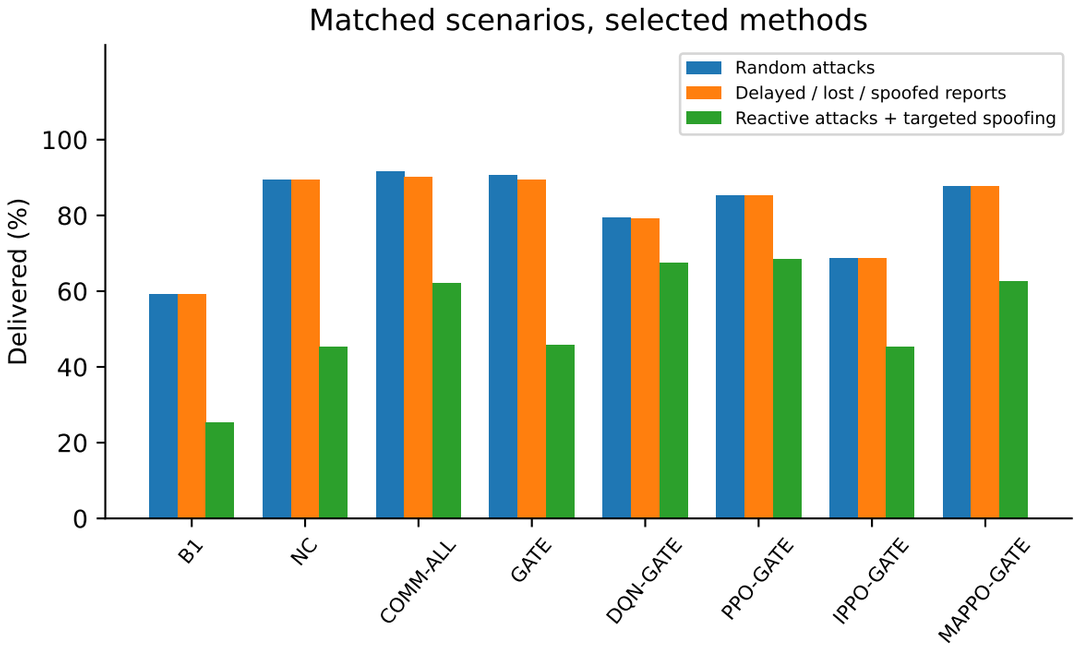
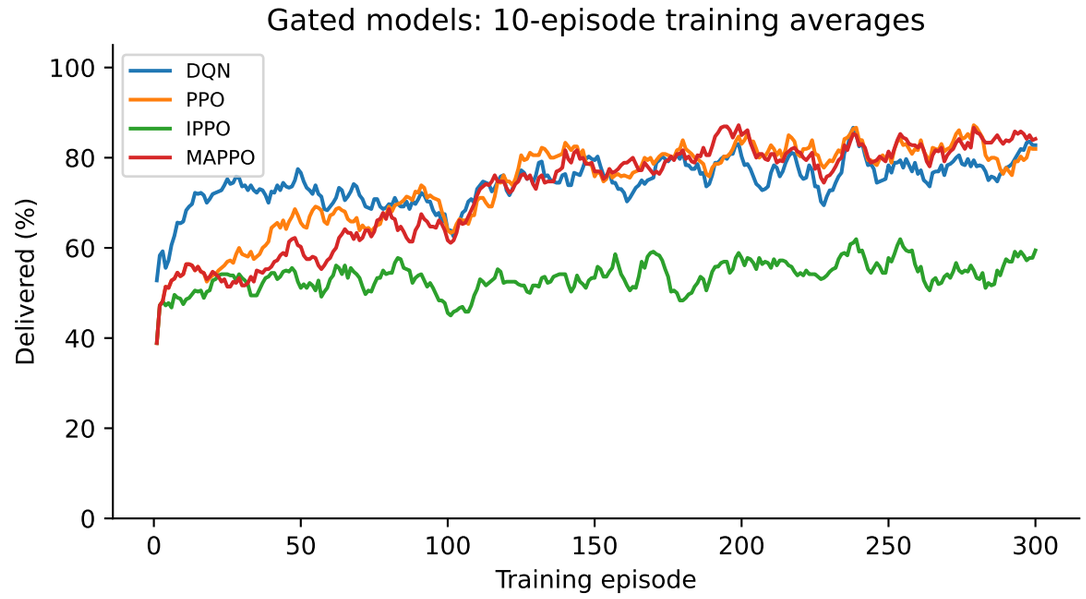
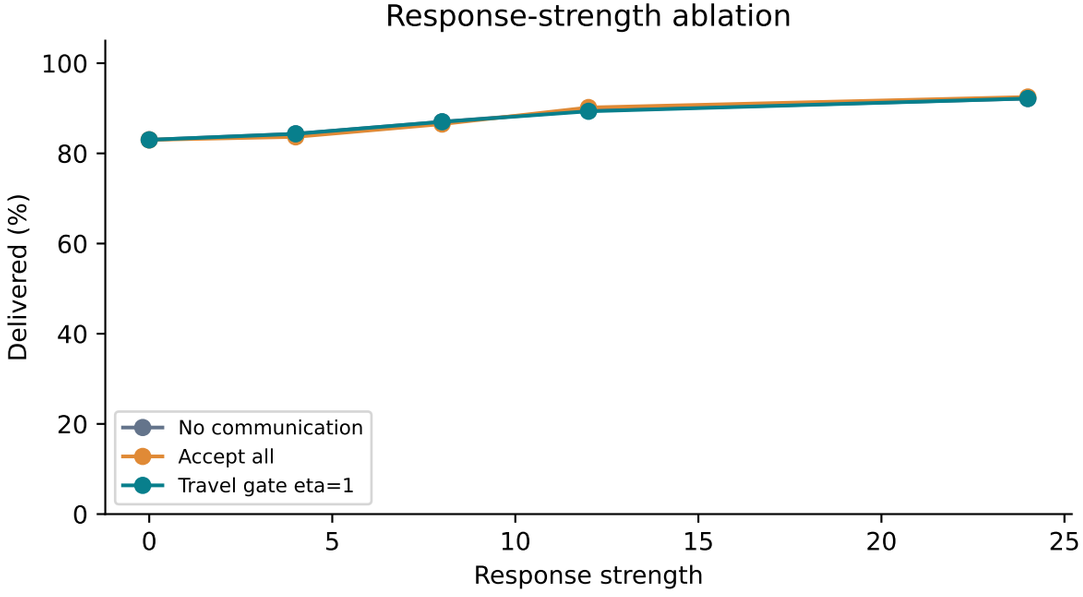
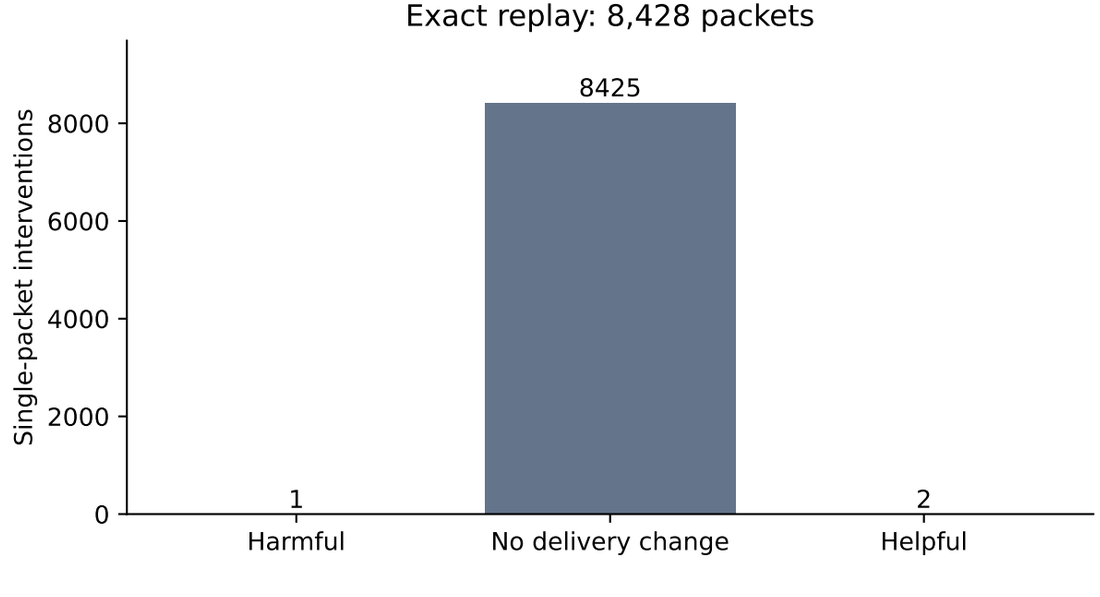
Updated results — longer training and harder threats





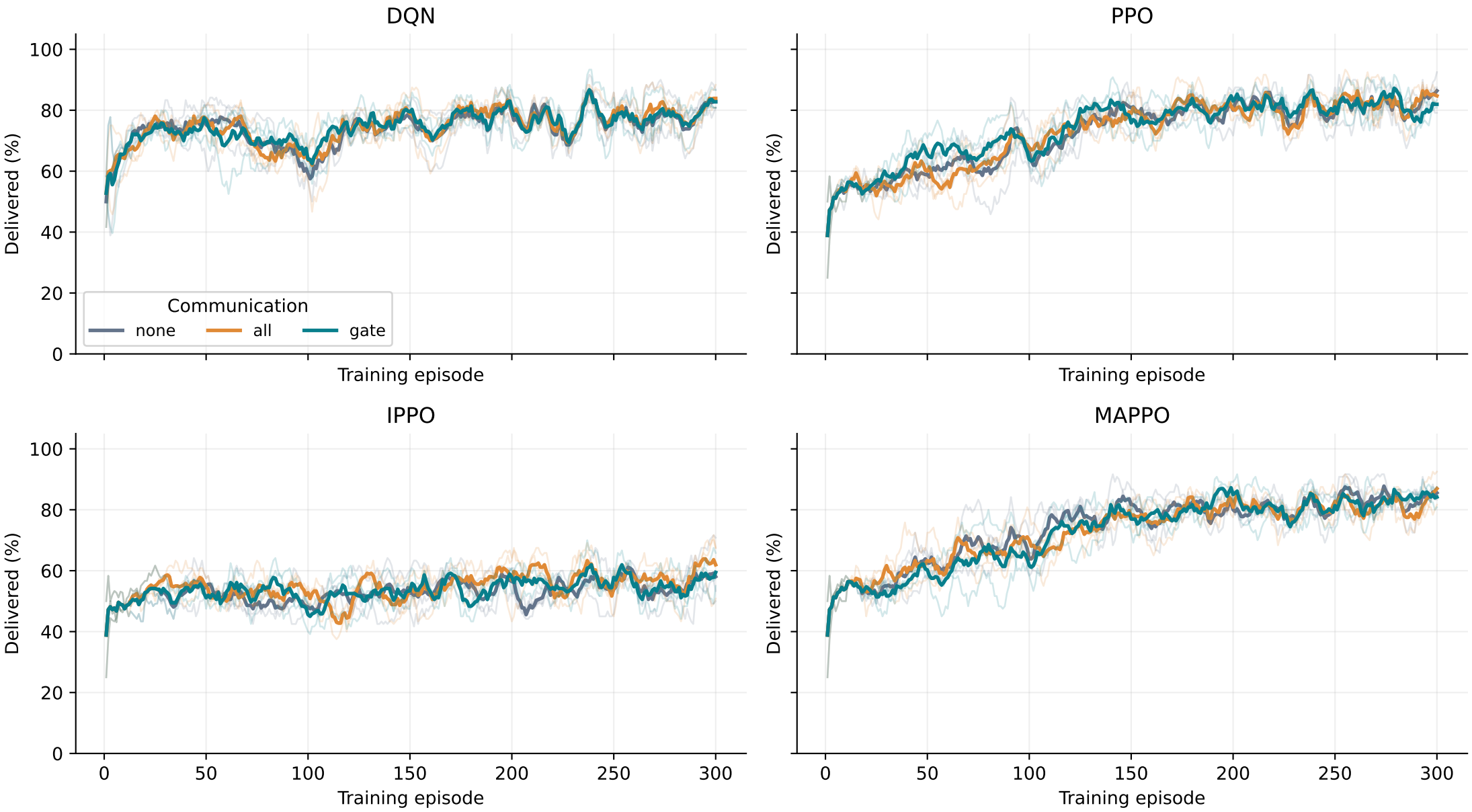




Three seeds, matched fresh held-out graphs, explicit graph uncertainty and independent gate/penalty ablation. Exploratory results, not a convergence claim.

EXPLORATORY RESULTS • 300 episodes/model • 3 training seeds • convergence not assumed

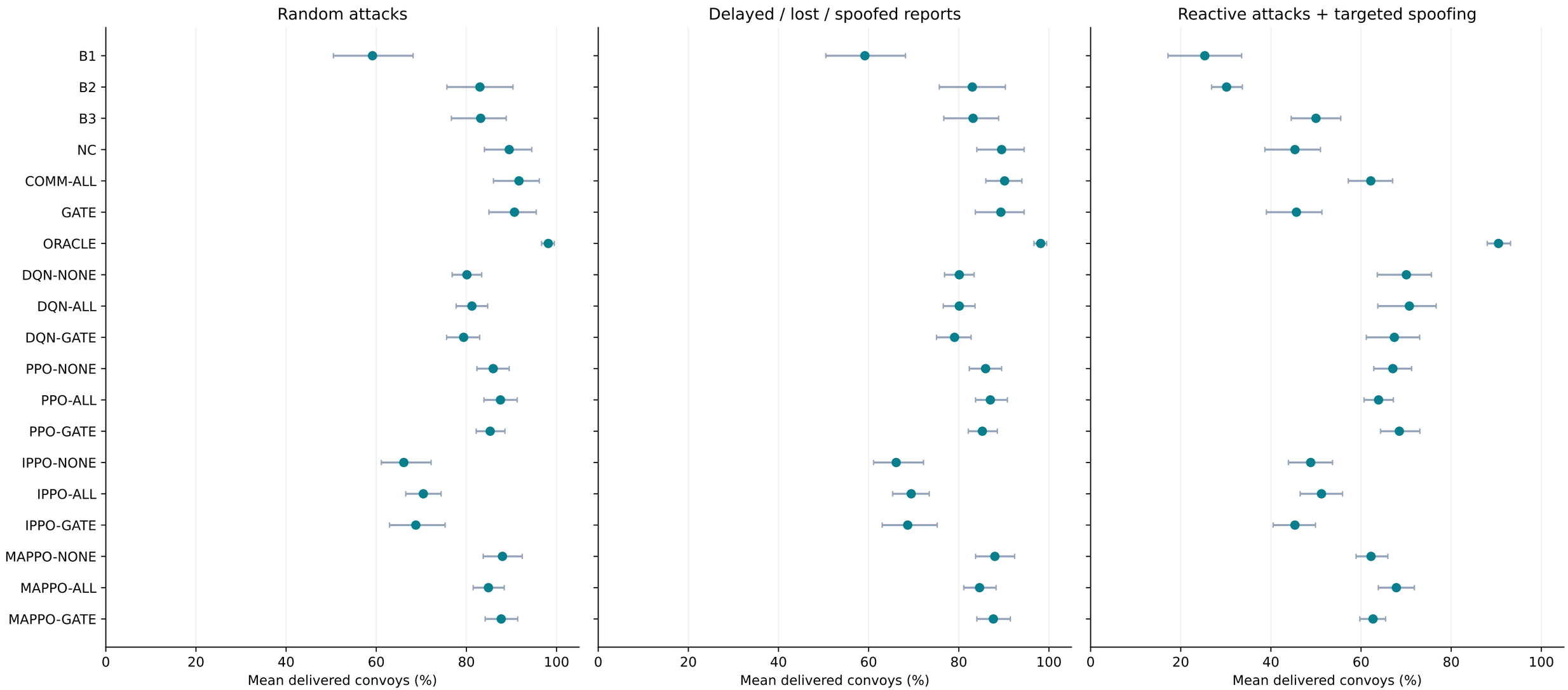
Longer training under contested conditions



10-episode moving averages. Bold: three-seed mean; faint: individual seeds. Stochastic training outcomes, not fixed-policy convergence tests.

EXPLORATORY RESULTS • 300 episodes/model • 3 training seeds • convergence not assumed

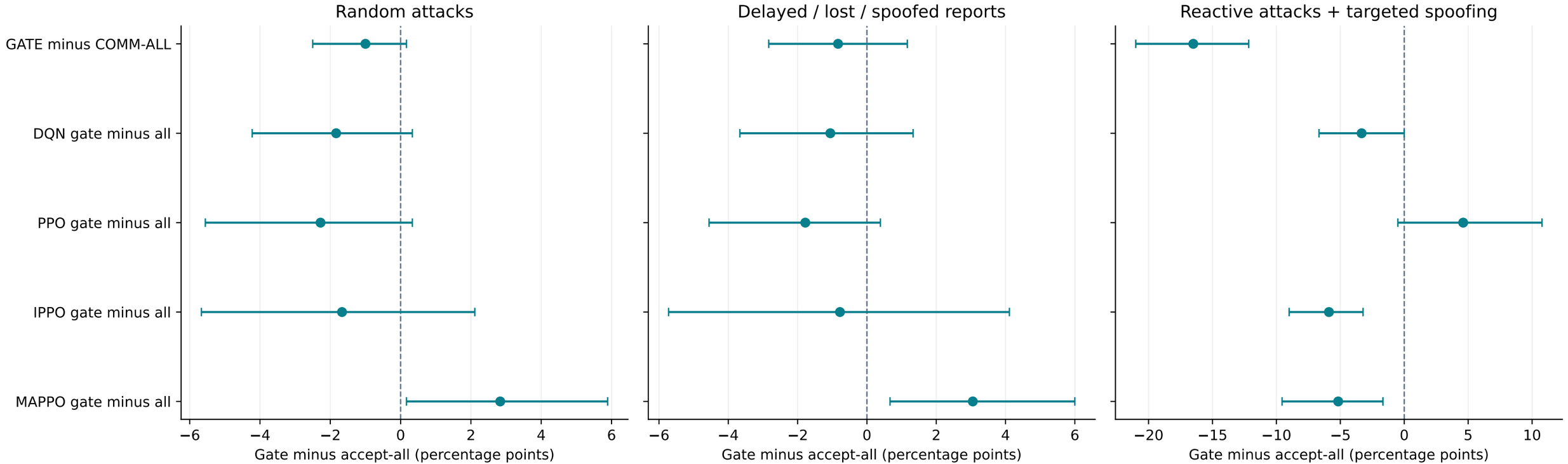
Matched held-out comparisons with uncertainty



Same 10 held-out graphs × 5 episodes across scenarios. Bars: graph-cluster 95% bootstrap intervals conditional on trained models; 12 convoys.

EXPLORATORY RESULTS • 300 episodes/model • 3 training seeds • convergence not assumed

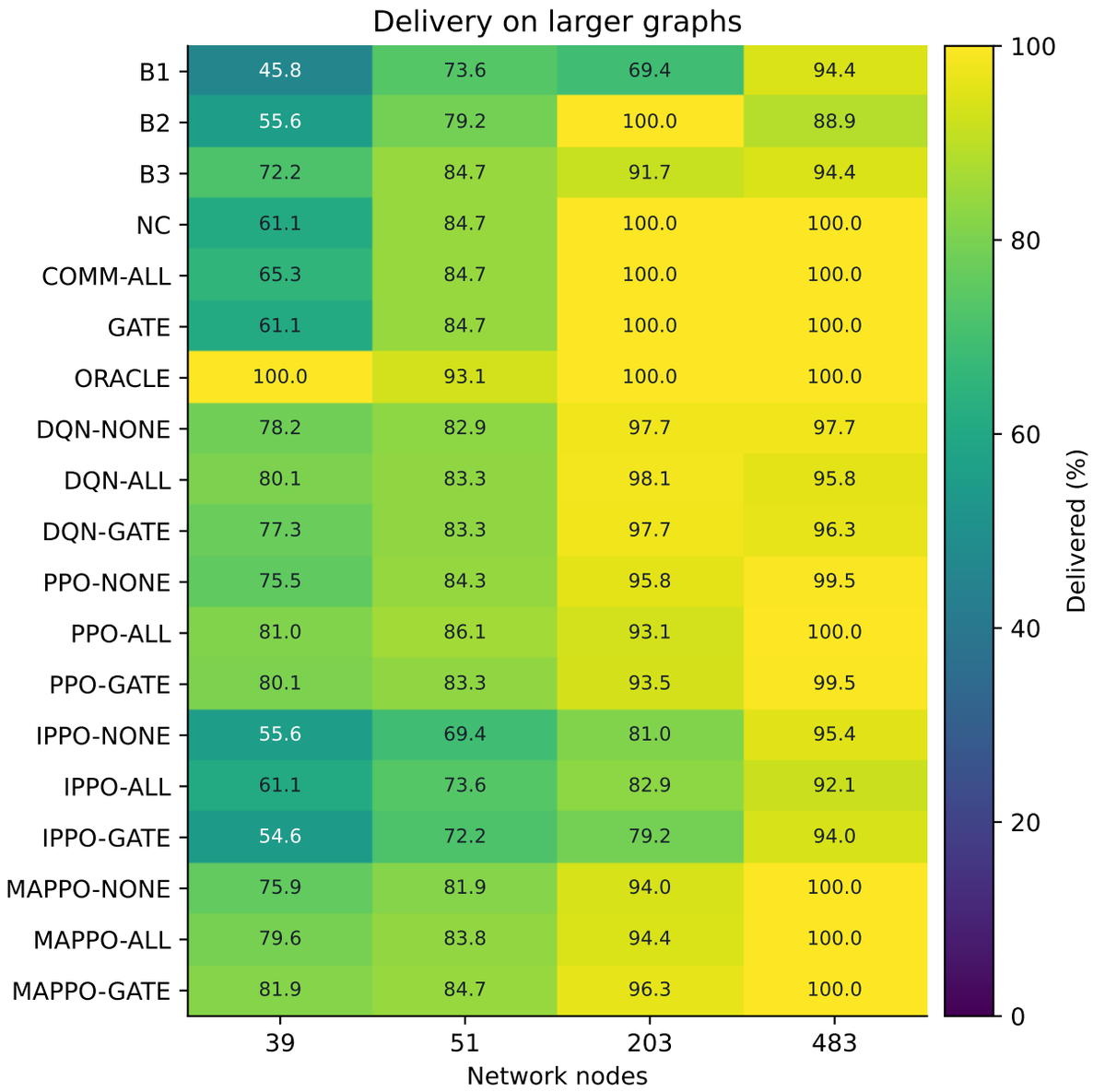
Does the gate improve delivery?

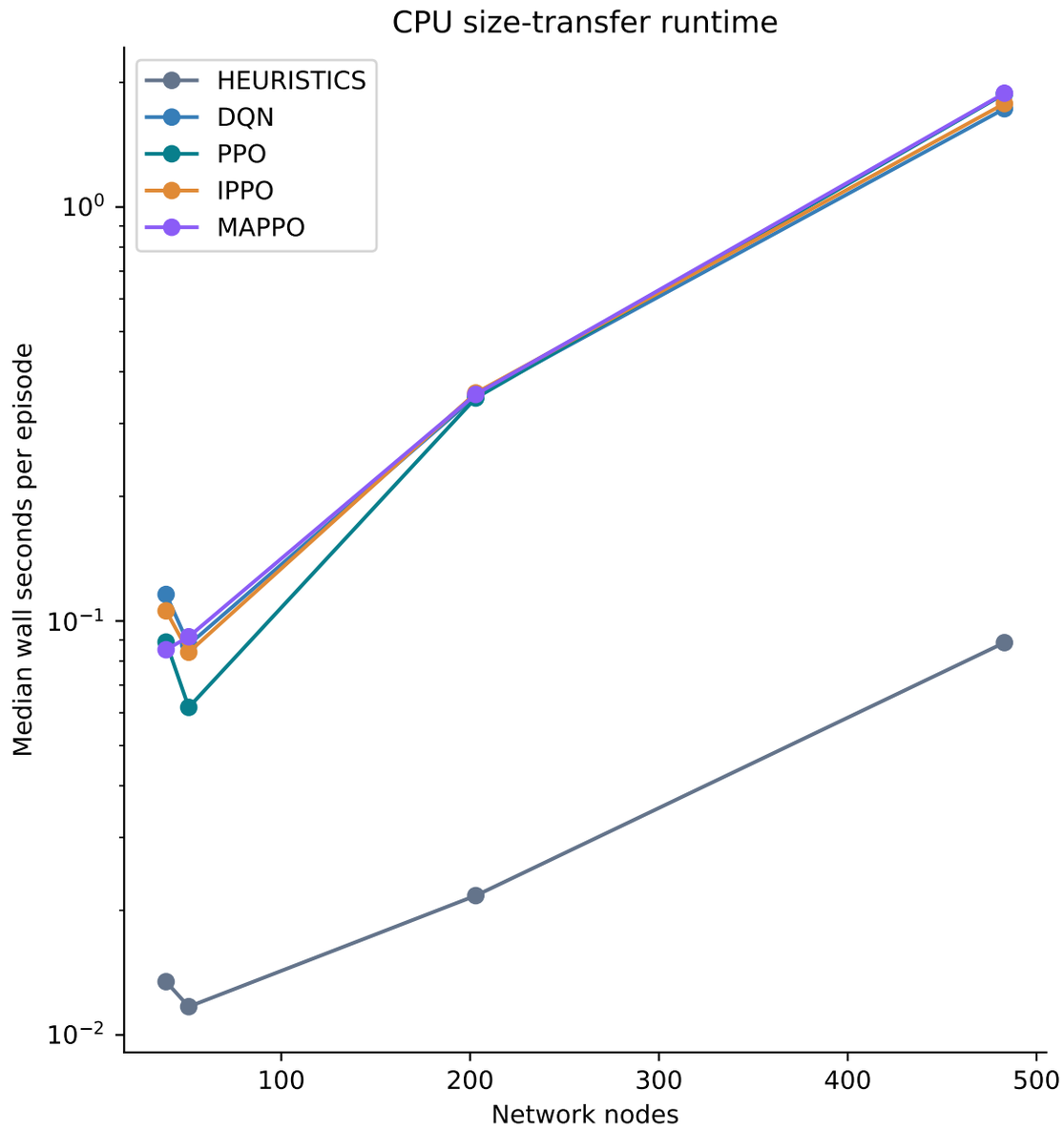


Paired 95% bootstrap intervals. Learned contrasts resample matched graph clusters and training seeds; zero-crossing intervals are inconclusive.

EXPLORATORY RESULTS • 300 episodes/model • 3 training seeds • convergence not assumed

Larger-network inference with newly trained models

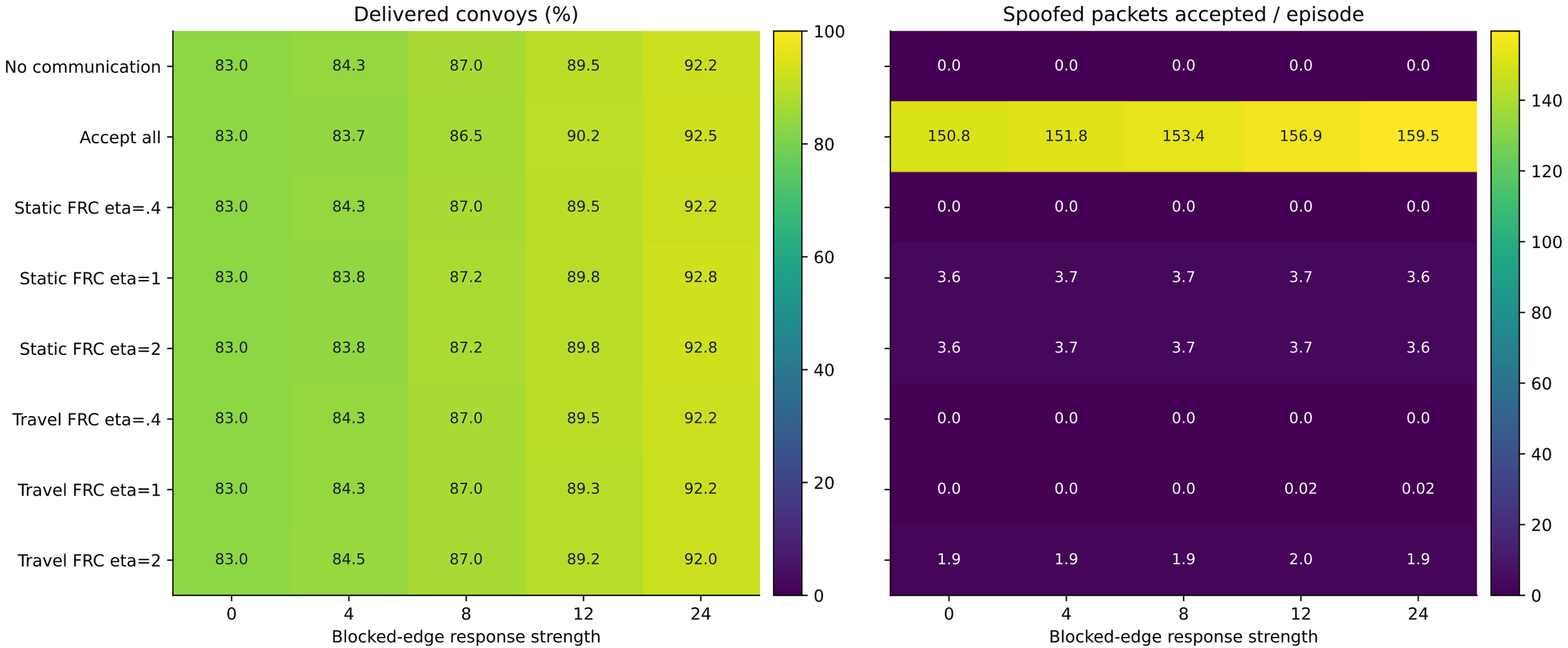




3 graphs × 2 episodes per size; population/attack budget fixed. Threat density therefore varies. Timing excludes Python allocation instrumentation.

EXPLORATORY RESULTS • 300 episodes/model • 3 training seeds • convergence not assumed

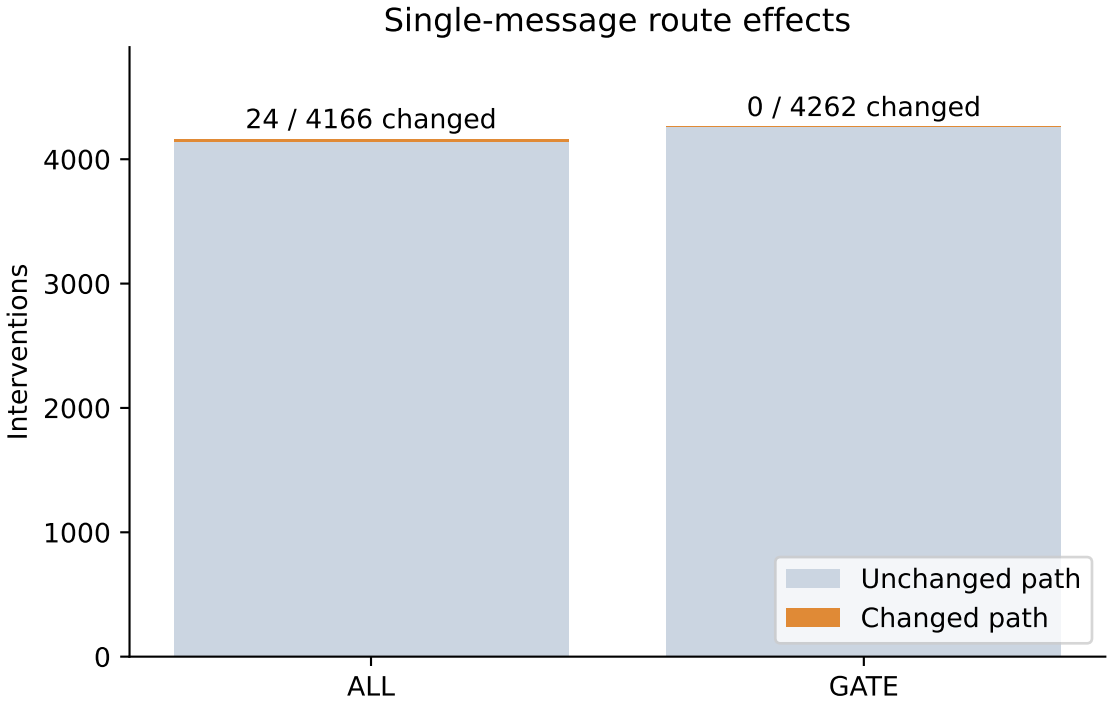
Gate filtering and response strength are separate factors

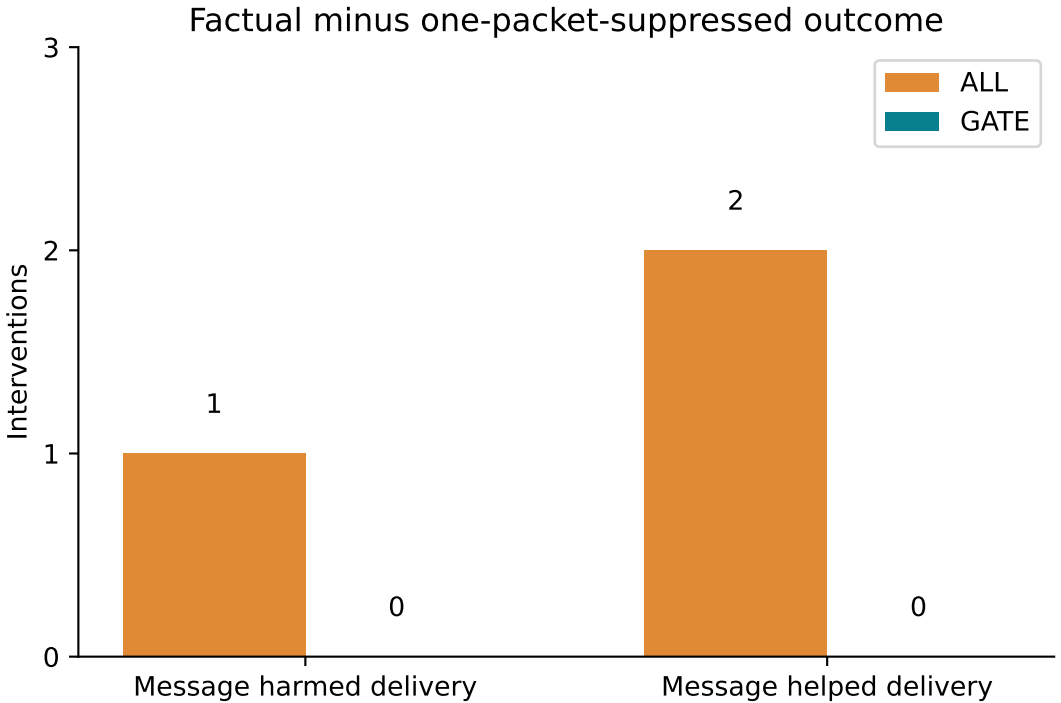


Delayed/lost/spoofed scenario; all methods share the same 10 graphs × 5 episodes. Full factorial data also includes every F/R/C switch combination.

EXPLORATORY RESULTS • 300 episodes/model • 3 training seeds • convergence not assumed

Exact single-message replay under stronger threats





8,428 interventions; all processed packets replayed, not sampled. Effects are realized counterfactuals; messages can have interacting effects.

EXPLORATORY RESULTS • 300 episodes/model • 3 training seeds • convergence not assumed